Homework

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1 Problem

Write a program that outputs all possible combinations of 0s and 1s.

Input an integer X, and output all possible combinations of 0 and 1 using exactly X bits.

2 Code Snippet

```c
#include <stdio.h>
#include <stdlib.h>

void rec(char bit[], int i, int n) {
  if (i == n) {
    printf("%s\n", bit);
    return;
  }

  bit[i] = '0';
  rec(bit, i + 1, n);
  bit[i] = '1';
  rec(bit, i + 1, n);
}

int main(void)
{
  printf("Input a number: ");
  int x; scanf("%d", &x);
  printf("\n");

  char *bit = malloc(sizeof(char) * (1 << x));

  rec(bit, 0, x);
}
```

3 Result

```
Input a number:
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111
```

4 Editorial

The problem requires generating all binary numbers from 0 to $(1 << x)-1$. Each bit can only be 0 or 1, so we can use backtracking to implement the solution and return when $i == x$.